

MATERIAL SAFETY DATA SHEET

1. PRODUCT AND MANUFACTURER IDENTIFICATION

CHLORFENAPYR 10% W/V SC

SUPPLIER NAME	Shandong Weifang Rainbow Chemical Co., Ltd.
SUPPLIER ADDRESS	Binhai Economic Development Area, Weifang, Shandong 262737, China
SUPPLIER PHONE	+86-531-88875230, 88875231
SUPPLIER FAX	+86-531-88875232, 88875224
CHEMICAL NAME	4-bromo-2-(4-chlorophenyl)-1-ethoxymethyl-5-trifluoromethyl-1H-pyrrole-3-carbonitrile
CAS NUMBER	122453-73-0
MOLECULAR FORMULA	C ₁₅ H ₁₁ BrClF ₃ N ₂ O
MOLECULAR WEIGHT	407.6
CHEMICAL FAMILY	pyrrole
USE	Insecticides

2. COMPOSITION / INFORMATION ON INGREDIENTS

CHLORFENAPYR	100G/L
OTHER NO HAZARD INGREDIENTS	up to 1L

3. HAZARDS IDENTIFICATION

Harmful by inhalation, in contact with skin and if swallowed.

Other information: Poisons schedule 6

4. FIRST - AID MEASURES

If swallowed: Call a poison control center or doctor immediately for treatment advice. Have person sip a glass of water if able to swallow. Do not induce vomiting unless told to by a poison control center or doctor.

Do not give anything to an unconscious person.

If on skin or clothing: Take off contaminated clothing. Rinse skin immediately with plenty of water for 15-20 minutes. Call a poison control center or doctor for treatment advice.

If inhaled: Move person to fresh air. If person is not breathing, call an ambulance, then give artificial respiration, preferably mouth-to-mouth if possible. Call a poison control center or doctor for further treatment advice.

If in eyes: Immediately hold eyelids open and flush with a steady, gentle stream of water for 15-20 minutes. Call a poison control center or doctor for treatment advice.

Note to physician: Treat symptomatically. No specific antidote.

Note: Have the product container or label with you when calling a poison control center or doctor or going for treatment.

5. FIRE FIGHTING MEASURES

Hazardous products of combustion: Non-combustible. However, following evaporation of aqueous component residual material can burn if ignited.

On burning may emit toxic fumes.

Extinguish fires with carbon dioxide, foam, dry chemical and water fog. Water spray can be used for cooling of unaffected stock. Use as little water as possible. Use spray or fog. Solid stream may cause spreading. Contain water used for fire fighting for later disposal. Avoid the accumulation of polluted run-off from the site.

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Fire fighting: Remove spectators from surrounding area. Isolate the fire area and evacuate downwind. Use a recommended extinguishing agent for the type of surrounding fire.

Fight fire from maximum distance and use unmanned hose holder or monitor nozzles. Contain fire control agents for later disposal. Avoid inhaling hazardous vapours and fumes from burning materials. Keep upwind.

Remove container from fire area if possible and without risk. Water can be used to cool unaffected containers but must be contained for later disposal. Use as little water as possible.

Dike fire control water for later disposal. Do not scatter the material. Avoid pollution of waterways. Do not use high volume water jet, due to contamination risk. Contain water used for fire fighting for later disposal. Avoid the accumulation of polluted run-off from the site.

Personal protective equipment: Fire fighters and others that may be exposed should wear full protective clothing and self-contained breathing apparatus in the positive-pressure mode with a full-face mask when there is a possibility of exposure.

Decontaminate emergency personnel with soap and water before leaving the fire area.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions: Take appropriate protective measures. Clear area. Shut off source of leak only under safe conditions.

Extinguish sources of ignition nearby and downwind. Ensure adequate ventilation. Wear suitable personal protective clothing and equipment.

Environmental precautions: Do not discharge into the subsoil/soil. Do not discharge into drains/surface waters/ground water. Contain contaminated water/firefighting water.

Clean up: Dike spillage. Pick up with suitable absorbent material. Place into suitable containers for reuse or disposal in a licensed facility. Spilled substance/product should be recovered and applied according to label rates whenever possible. If application of spilled substance/product is not possible, then spills should be contained, solidified, and placed in suitable containers for disposal. After decontamination, spill area can be washed with water. Collect wash water for approved disposal.

7. HANDLING AND STORAGE

Handling:

Harmful if inhaled or swallowed. Irritating to eyes and skin.

Avoid contact with eyes and skin, and inhalation of spray and vapour. Use with adequate ventilation. Wash hands before eating, drinking, chewing gum, smoking, or using the toilet. Operators should change and wash clothing daily. Remove clothing immediately if the pesticide gets inside. Then wash skin thoroughly using a non-abrasive soap and put on clean clothing. Do not apply directly to areas where surface water is present, or to intertidal areas below the mean high water mark. Water used to clean equipment must be disposed of correctly to avoid contamination.

Storage:

Keep under lock and key and out of reach of unauthorized persons, children and animals. Store in its original labeled container in isolated, dry, cool and well-ventilated area.

Chlorfenapyr becomes unstable on prolonged storage above 37 °C and is rapidly decomposed by UV. Not to be stored close to foodstuffs, animal feeds and water supplies. Local regulations should be complied with.

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

Respiratory Protection: Supplied air respirators should be worn if large quantities of mist are generated or prolonged exposure possible.

Eye Protection: Chemical goggles when respirator does not provide eye protection.

Protective Clothing: Gloves and protective clothing as necessary to prevent skin contact.

Ventilation: Whenever possible, engineering controls should be used to minimize the need for personal protective equipment.

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9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	Off-White flowable suspension
pH:	6.0-9.0
Density:	1.05 g/cm ³
Flammability:	Non-flammable
Explosion properties:	Not explosive

10. STABILITY AND REACTIVITY

STABILITY: Stable. Do not store below 32° F. Avoid heat and sunlight.

CONDITIONS TO AVOID: Store in original container in cool, dry, well ventilated place away from ignition sources, heat or flame.

CHEMICAL INCOMPATIBILITY: Oxidizing agents.

HAZARDOUS DECOMPOSITION PRODUCTS: Including oxides of carbon and nitrogen; HCl; HF; HBr

HAZARDOUS POLYMERIZATION: Does not occur.

CONDITIONS TO AVOID: Does not polymerize.

11. TOXICOLOGICAL INFORMATION

ACUTE TOXICITY:

ORAL LD50 (RAT):	MALES: >900 mg/Kg, FEMALES: >900 mg/kg
DERMAL LD50 (RABBIT):	MALES: >3500 mg/Kg, FEMALES: >3500 mg/kg
INHALATION (RAT):	MALES: > 2.4 mg/L, FEMALES: > 2.4 mg/L
EYE IRRITATION (RABBIT):	Moderate eye irritation
SKIN IRRITATION (RABBIT):	Moderate dermal irritant
SKIN SENSITIZATION (GUINEA PIGS):	Not a dermal sensitizer

12. ECOLOGICAL INFORMATION

Mobility, Degradability & Accumulation: Animals In rats, >60 % of orally administered Chlorfenapyr was excreted primarily through faeces within 24 hours. The absorbed residues were metabolised via N-dealkylation, dehalogenation, hydroxylation and conjugation. Parent and less polar metabolites were found in egg, milk and tissues such as fat and liver. Metabolism in hens and goats is similar to that in rats, however in these species, 80 % of orally administered Chlorfenapyr was rapidly excreted. Unexcreted residues were present in kidney and liver. At the potential maximum dietary burden, all residues are <0.01 ppm. Chlorfenapyr is the only significant residue component.

ECOTOXICOLOGY:

Earthworm: NOEC (14 d) for Eisenia foetida 8.4 mg/kg.

Daphnia: LC50 (96 h) 6.11 µg/l.

Algae: EC50 for Selenastrum capricornutum 132 ppb.

13. DISPOSAL CONSIDERATIONS

Products Disposal: on site disposal of the concentrated product is not acceptable. Ideally, the product should be used for its intended purpose. If there is a need to dispose of the product, approach local authorities who hold periodic collections of unwanted chemicals.

Container disposal: Triple rinse containers, add rinsate to the spray tank, then offer the container for recycling/reconditioning, or puncture top, sides and bottom and dispose of in landfill in accordance with local regulation.

Returnable containers: empty contents fully into application equipment. Replace cap, close all valves and return to the point of supply for refill or

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storage.

If no landfill is available, bury the containers below 500 mm in a disposal pit specifically marked and set up for this purpose clear of waterways, desirable vegetation and tree roots.

Empty containers and product should not be burnt.

14. TRANSPORT INFORMATION

Hazard class or division: 9

UN numbers: 3082

Packing Group: III

15. OTHER INFORMATION

The data given here is based on current knowledge and experience. The purpose of this safety data sheet is to describe the products in terms of their safety requirements. The data does not signify any warranty with regard to the products' properties.

ORIGINAL CONTROLLED